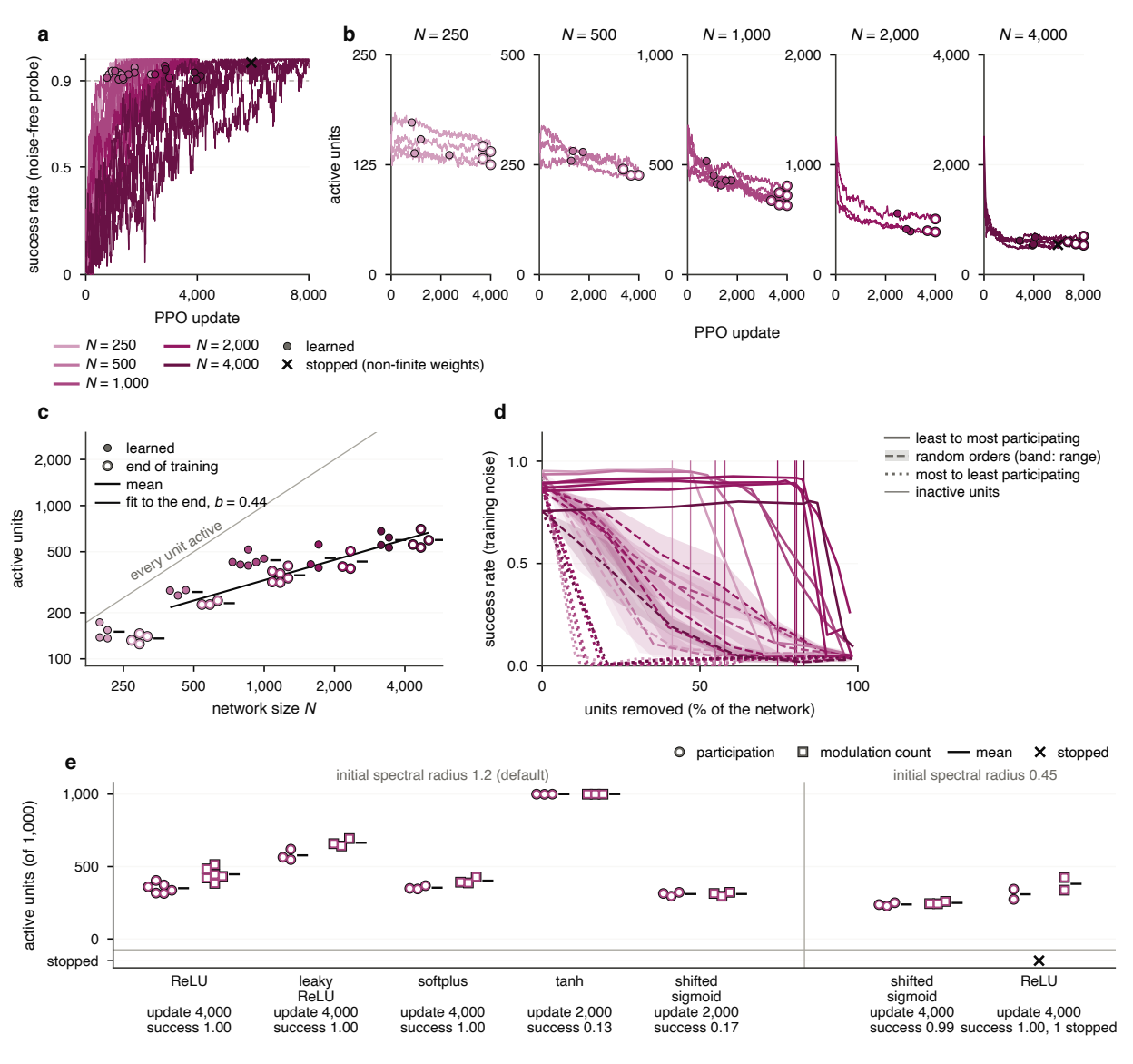

a
success rate (noise-free probe)
0
0.5
0.9
0
4,000
8,000
PPO update
b
N = 250
N = 500
N = 1,000
N = 2,000
N = 4,000
active units
250
125
0
500
250
1,000
500
2,000
1,000
4,000
2,000
PPO update
N = 250
N = 500
N = 1,000
N = 2,000
N = 4,000
learned
stopped (non-finite weights)
c
learned
end of training
mean
fit to the end, b = 0.44
every unit active
active units
2,000
1,000
500
200
100
250
500
1,000
2,000
4,000
network size N
d
success rate (training noise)
1.0
0.5
0.0
0
50
100
units removed (% of the network)
least to most participating
random orders (band: range)
most to least participating
inactive units
e
participation
modulation count
mean
stopped
initial spectral radius 1.2 (default)
initial spectral radius 0.45
active units (of 1,000)
1,000
500
0
stopped
ReLU
update 4,000
success 1.00
leaky ReLU
update 4,000
success 1.00
softplus
update 4,000
success 1.00
tanh
update 2,000
success 0.13
shifted sigmoid
update 2,000
success 0.17
shifted sigmoid
update 4,000
success 0.99
ReLU
update 4,000
success 1.00, 1 stopped